

Train on sidebands

Predict b and C_{eff}
Factor $C_{\text{eff}} = LL^T$



Test one yield

$\lambda = b + L\theta + Aw$
Hold b, L, w fixed



Profile the background

Minimize Poisson NLL
plus $\frac{1}{2}\theta^T\theta$



Build the profile

Record $\ell_p(A)$
Fit $\hat{A}$ or test A_{90}

$L\theta$ moves bins coherently; the Gaussian penalty controls the size of that deformation.